



Dear Dr. Way,

Please consider our paper, “Warmer and longer growing seasons increase growth for juvenile trees in a common garden” for publication as a research article in *Global Change Biology*.

What scientific question is addressed in this manuscript? Do trees grow more with longer, warmer seasons? To answer this question, we used a unique common garden dataset of radial growth data from juvenile trees of four species sourced from multiple provenances. Using paired leaf phenology data on budburst and budset we tested the effect of both longer and warmer seasons.

What is/are the key finding(s) that answer this question? We found that warmer seasons increased tree growth, with thermal season being a better predictor for growth than calendar season. We suggest later season warmth may be more important to growth, which is supported by later budset driving increased growth and a little role of an earlier leafout.

Why is this work important and timely? Our work brings important insights on how longer, warmer seasons affect tree growth. Recent work has challenged the assumption that longer growing seasons drive increased tree growth, with major implications for the future of temperate forest carbon sink in a warming climate (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023).

Describe how your paper fits within the scope of GCB. Our work provides new evidence about how trees grow in response to longer and warmer seasons, with implications for carbon storage and forest dynamics. We show that young deciduous trees may actually thrive under a warmer climate, which could affect forest regeneration and range expansion with warming (Luu *et al.*, 2024).

What biological AND global change aspects does it address? Our paper addresses fundamental questions about how tree grow, from the angle of how climate change will alter growing seasons and thus tree growth. Our results suggest the role of late-season events and the thermal season may be critical to understanding and predicting tree growth in the future.

What are the three most recently published papers that are relevant to this manuscript?
Dow *et al.* (2022), (Silvestro *et al.*, 2023), (Wolkovich *et al.*, 2025)

Provide the name and institutional email addresses of five potential reviewers.

Caroline Polgar—carolinepolgar@gmail.com

Amanda Gallinat—agallina@colby.edu

Libby Ellwood—ellwood@floridamuseum.ufl.edu

Loïc D’Orangeville—loic.dorangeville@sbf.ulaval.ca

Julia Green—juliakgreen@arizona.edu

All authors contributed to this work and approved this version for submission. The manuscript is 3977 words with a 2820-word abstract, and 3 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *Global Change Biology*, and look forward to hearing from you.

Sincerely,

Christophe Rouleau-Desrochers

MSc in Forestry

Department of Forest & Conservation Sciences

The University of British Columbia

- Dow C, Kim AY, D'Orangeville L, *et al.* (2022) Warm springs alter timing but not total growth of temperate deciduous trees. *Nature*, **608**, 552–557. doi:10.1038/s41586-022-05092-3. URL <https://www.nature.com/articles/s41586-022-05092-3>.
- Luu H, Ris Lambers JH, Lutz JA, Metz M, Snell RS (2024) The importance of regeneration processes on forest biodiversity in old-growth forests in the Pacific Northwest. *Philosophical Transactions of the Royal Society B: Biological Sciences*, **379**, 20230016. doi:10.1098/rstb.2023.0016. URL <http://royalsocietypublishing.org/rstb/article/42852>.
- Matula R, Knířová S, Vítámvás J, *et al.* (2023) Shifts in intra-annual growth dynamics drive a decline in productivity of temperate trees in Central European forest under warmer climate. *Science of The Total Environment*, **905**, 166906. doi:10.1016/j.scitotenv.2023.166906. URL <https://linkinghub.elsevier.com/retrieve/pii/S0048969723055316>.
- Silvestro R, Zeng Q, Buttò V, *et al.* (2023) A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports*, **13**, 4059. doi:10.1038/s41598-023-31336-x. URL <https://www.nature.com/articles/s41598-023-31336-x>.
- Wolkovich EM, Ettinger AK, Chin AR, *et al.* (2025) Why longer seasons with climate change may not increase tree growth. *Nature Climate Change*, **15**, 1283–1292. doi:10.1038/s41558-025-02476-x. URL <https://www.nature.com/articles/s41558-025-02476-x>.